FETAL PERIOD

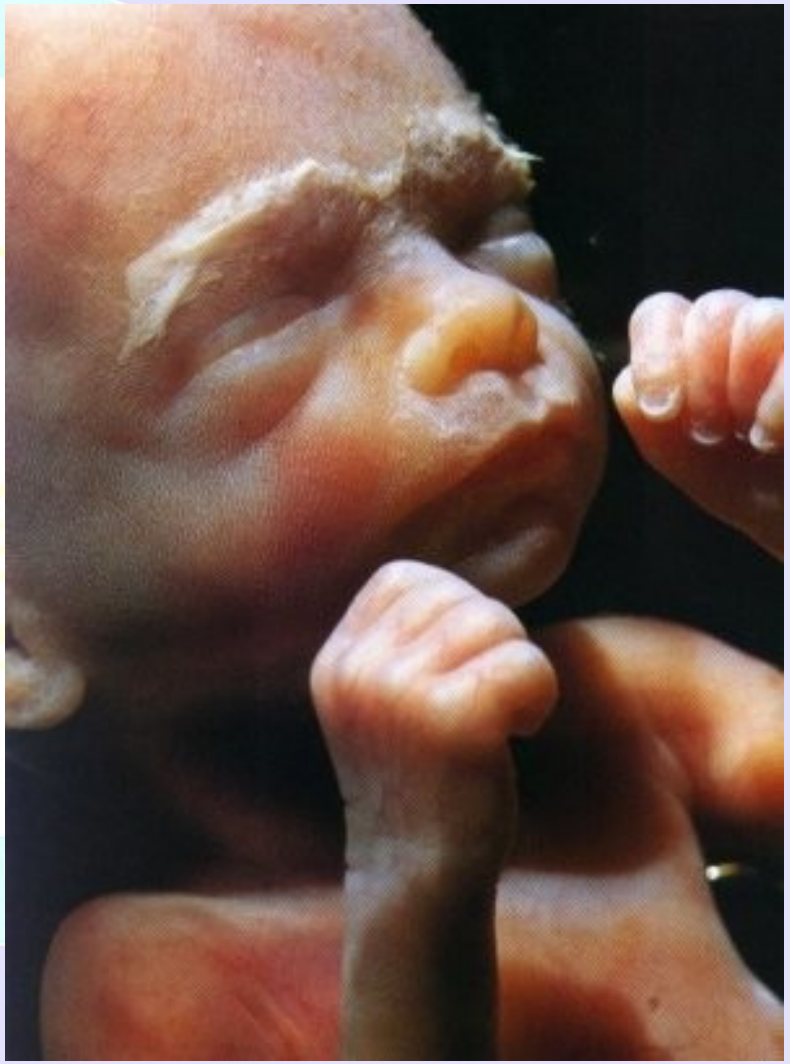
(9th WEEK TILL BIRTH)

LEARNING OBJECTIVE

- At the end of lecture student should be able to,
- Define what is fetal period?
- Discus the factors affecting fetal period.
- Specify the Tissue and organ development week wise.
- Know the different milestone in development of fetus.
- Mention the cause of fetal loss.

FETAL PERIOD

- The fetal period begins at the end of the 10th week of gestation (8th week of development).
- From the 8th week until birth (around 38 weeks), the developing organism is called a fetus.



FETAL PERIOD

The fetal period is characterized by **two** processes.

- The first is rapid growth (increase in size and cell number).
- The second is continued tissue and organ differentiation (specialization of cells to perform distinct functions).
- The third is relative slow down in the growth of the head compare with the rest of the body.

FACTORS AFFECT FETAL GROWTH



Many factors affect fetal growth,

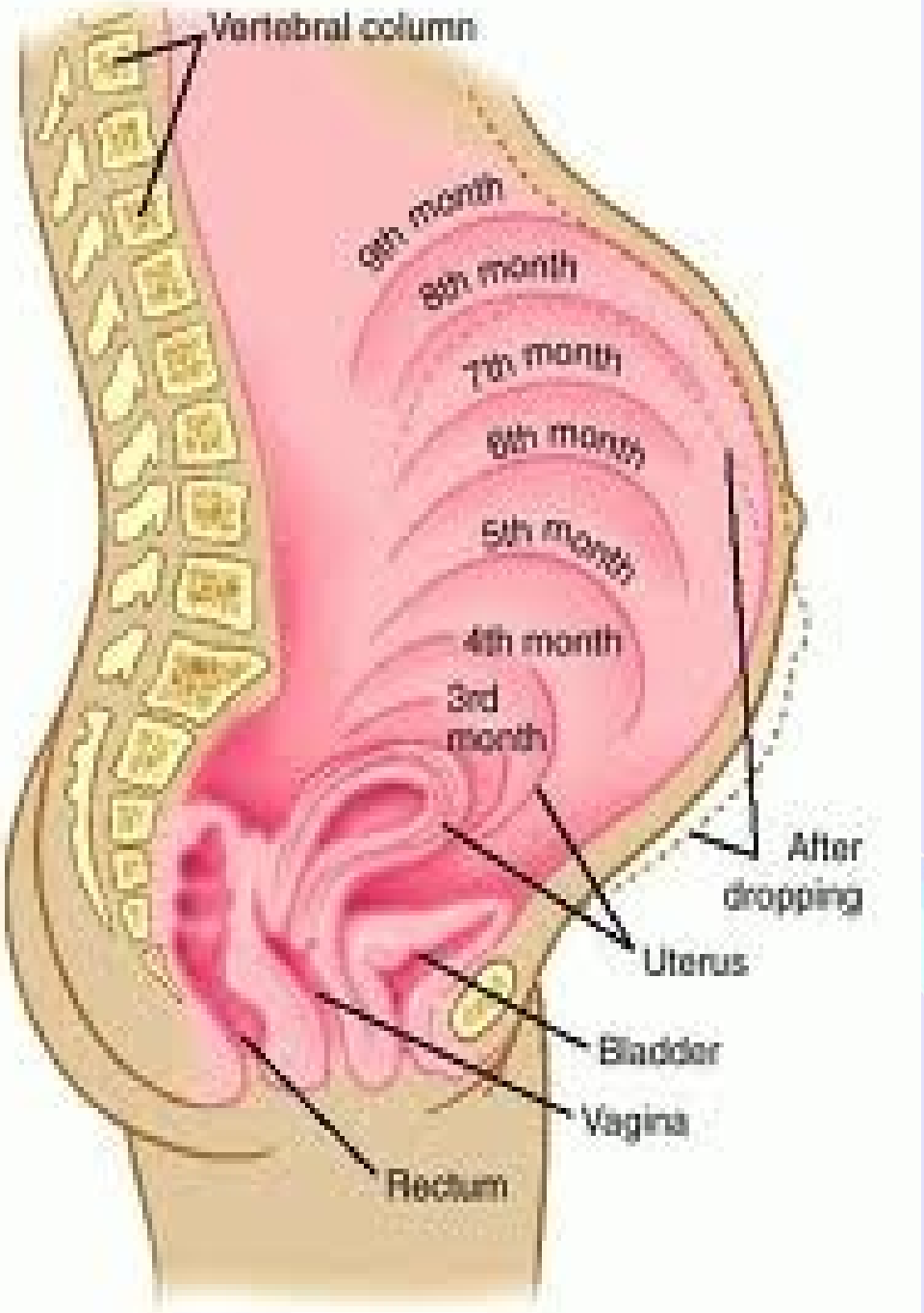
- The nutritional state and social habits (e.g., smoking, drug use) of the mother.
- The state of placental function.
- The genetic makeup of the fetus



RAPID GROWTH

- Fetal growth rate is greatest at the beginning of the fetal period (through week sixteen), during which time the fetus increases twenty-five-fold in weight.
- The largest increase in absolute weight gain, however, takes place during the final month of gestation.

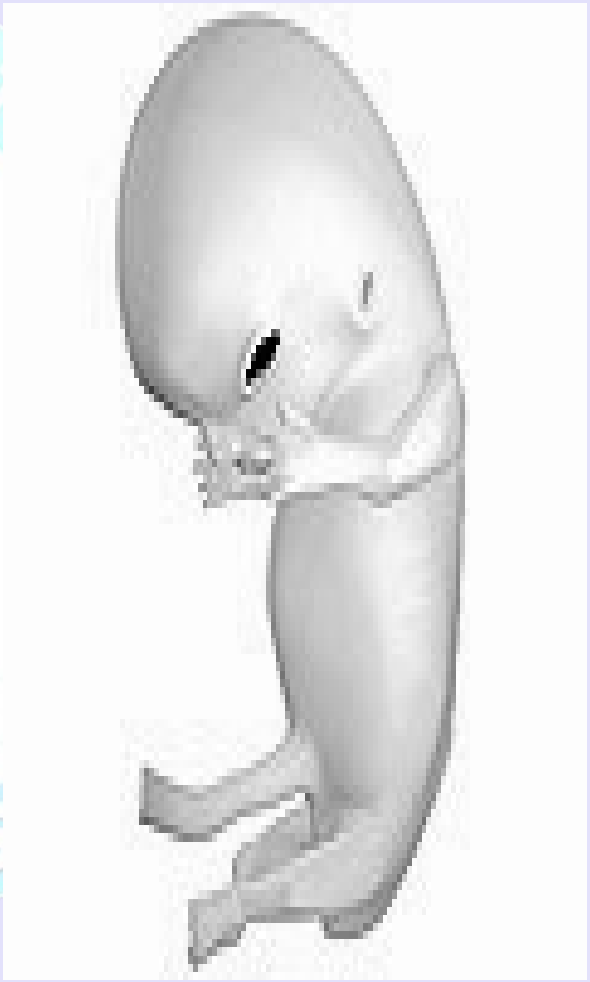



Vertebral column
9th month
8th month
7th month
6th month
5th month
4th month
3rd month
After dropping
Uterus
Bladder
Vagina
Rectum

TISSUE AND ORGAN DEVELOPMENT

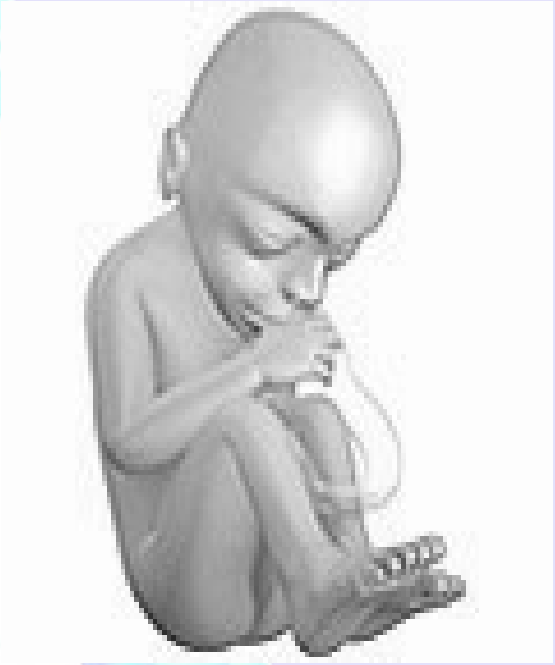
WEEKS 10–12

- Gestational age: 9–11 weeks old.
- Embryonic age: Weeks number 8–10. 7–9 weeks old.
- Embryo measures 30–80 mm (1.2–3.2 inches) in length.
- Ventral and dorsal pancreatic buds fuse during the 8th week
- Intestines rotate.
- Facial features continue to develop.
- The eyelids are more developed.
- The external features of the ear begin to take their final shape.
- The head comprises nearly half of the fetus' size.
- The face is well formed
- The eyelids close and will not reopen until about the 28th week.
- Tooth buds, which will form the baby teeth, appear.
- The limbs are long and thin.
- The fetus can make a fist with its fingers.
- Genitals appear well differentiated.
- Red blood cells are produced in the liver.



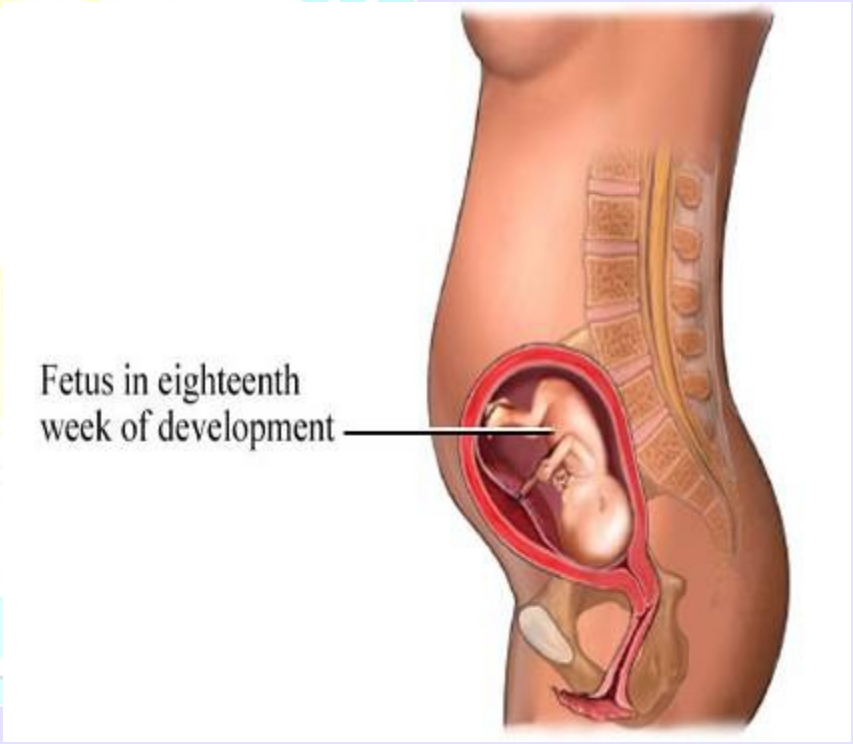
Weeks 13 to 16

- Gestational age: 12–15 weeks old.
- Embryonic age: Weeks number 11–14. 10–13 weeks old.
- The fetus reaches a length of about 15 cm (6 inches).
- A fine hair called lanugo develops on the head.
- Fetal skin is almost transparent.
- More muscle tissue and bones have developed, and the bones become harder.
- The fetus makes active movements.
- Sucking motions are made with the mouth.
- Meconium is made in the intestinal tract.
- The liver and pancreas produce fluid secretions.
- From week 13, sex prediction by obstetric ultrasonography is almost 100% accurate.
- At week 15, main development of external genitalia is finished



WEEK 19

- Gestational age: 18 weeks old.
- Embryonic age: Week numbers 17. 16 weeks old.
- The fetus reaches a length of 20 cm (8 inches).
- Lanugo covers the entire body.
- Eyebrows and eyelashes appear.
- Nails appear on fingers and toes.
- The fetus is more active with increased muscle development.
- "Quickening" usually occurs (the mother and others can feel the fetus moving).
- The fetal heartbeat can be heard with a stethoscope.



WEEK 23

- Gestational age: 22 weeks old.
- Embryonic age: Week numbers 21. 20 weeks old.
- The fetus reaches a length of 28 cm (11.2 inches).
- The fetus weighs about 725 g (1 lb 10 oz).
- Eyebrows and eyelashes are well formed.
- All of the eye components are developed.
- The fetus has a hand and startle reflex.
- Footprints and fingerprints continue forming.
- Alveoli (air sacs) are forming in lungs.



WEEK 27

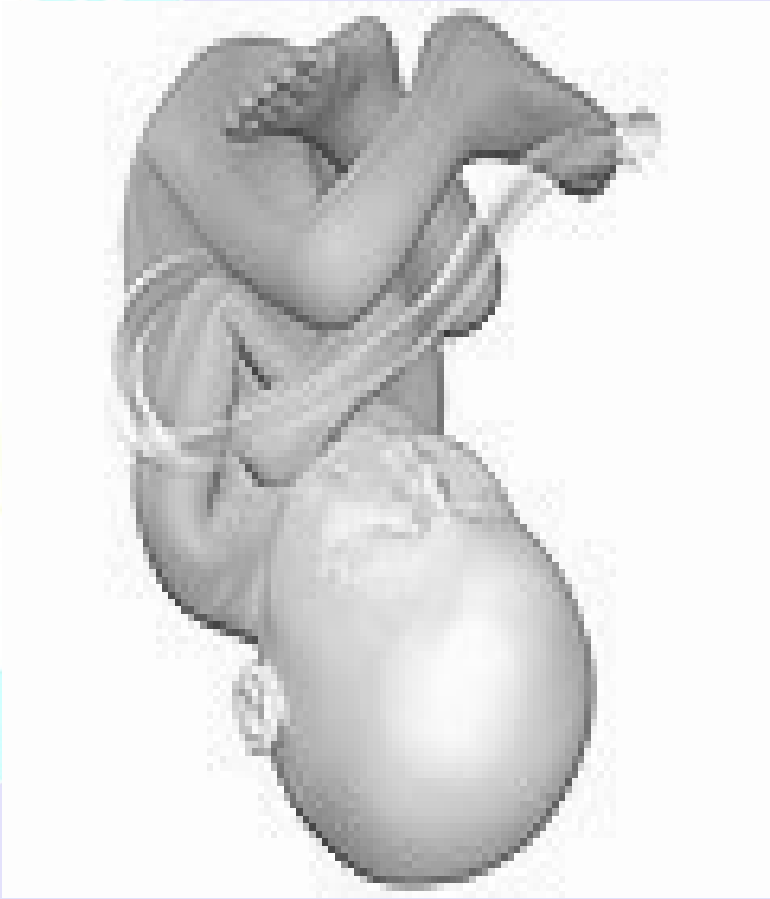
- Gestational age: 26 weeks old.
- Embryonic age: Week numbers 25. 24 weeks old.
- The fetus reaches a length of 38 cm (15 inches).
- The fetus weighs about 1.2 kg (2 lb 11 oz).
- The brain develops rapidly.
- The nervous system develops enough to control some body functions.
- The eyelids open and close.
- The cochleae are now developed, though the myelin sheaths in neural portion of the auditory system will continue to develop until 18 months after birth.
- The respiratory system, while immature, has developed to the point where gas exchange is possible.

WEEK 31

- Gestational age: 30 weeks old.
- Embryonic age: Week numbers 29. 28 weeks old.
- The fetus reaches a length of about 38–43 cm (15–17 inches).
- The fetus weighs about 2 kg (3 lb 0 oz).
- The amount of body fat rapidly increases.
- Rhythmic breathing movements occur, but lungs are not fully mature.
- Thalamic brain connections, which mediate sensory input, form.
- Bones are fully developed, but are still soft and pliable.
- The fetus begins storing iron, calcium, and phosphorus.

WEEK 35

- Gestational age: 34 weeks old.
- Embryonic age: Week numbers 33. 32 weeks old.
- The fetus reaches a length of about 40–48 cm (16–19 inches).
- The fetus weighs about 2.5 to 3 kg (5 lb 12 oz to 6 lb 12 oz).
- Lanugo begins to disappear.
- Body fat increases.
- Fingernails reach the end of the fingertips.
- A baby born at 36 weeks has a high chance of survival, but may require medical interventions.



WEEKS 36 to 39

- Gestational age: 35–38 weeks old.
- Embryonic age: Weeks numbers 34–37. 33–36 weeks old.
- The fetus is considered full-term at the end of the 37th week of gestational age.
- It may be 48 to 53 cm (19 to 21 inches) in length.
- The lanugo is gone except on the upper arms and shoulders.
- Fingernails extend beyond fingertips.
- Small breast buds are present on both sexes.
- Head hair is now coarse and thickest.

BIRTH

The process of birth, or parturition (labor), occurs in three stages.

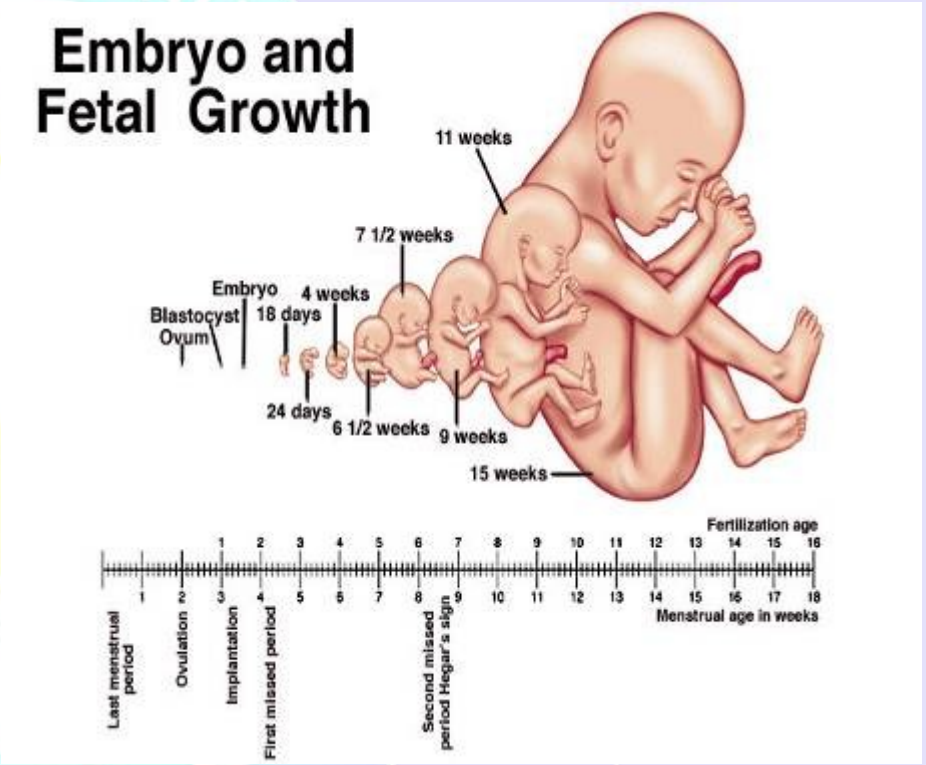
- The first is the **dilation** of the cervix.
- The second is the actual delivery of the fetus.
- The third ends with the expulsion of the placenta.

The entire process may take from only a few hours to well over a day to complete.

- The fetus is not as sensitive to damage from environmental exposures as the embryo, and toxic exposures often cause physiological abnormalities or minor congenital malformation.
- All major structures are already formed in the fetus, but they continue to grow and develop.

FETAL PERIOD GROWTH

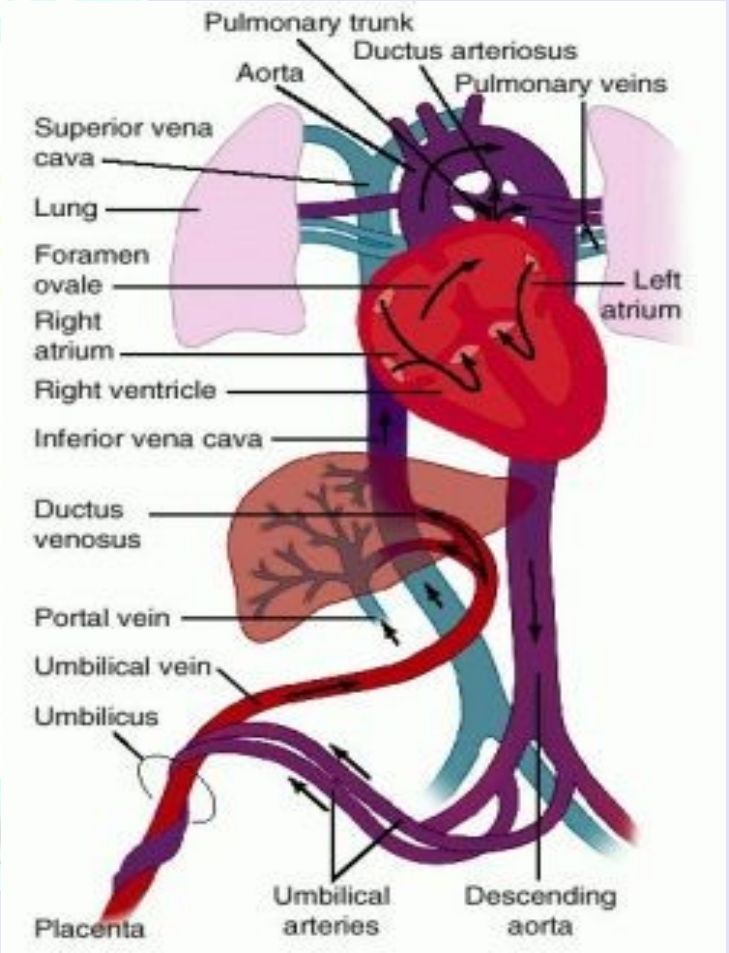
- The fetus grows dramatically in length and weight during the fetal period.
- The rest of the body grows more rapidly than the head.
- So the relative size of the head shrinks during the fetal period from roughly ½ to ¼ of the crown-heel length.



THE MILESTONES OF THE FETAL PERIOD

- sounds can be detected as early as 8 or 9 weeks post fertilization (p.f.) with a fetal Doppler ultrasound stethoscope, and 20 weeks post fertilization with a stethoscope or fetoscope.
- An external electronic fetal monitor can be used to obtain fetal heart rate in the third trimester in antepartum testing and intrapartum monitoring.

HEART



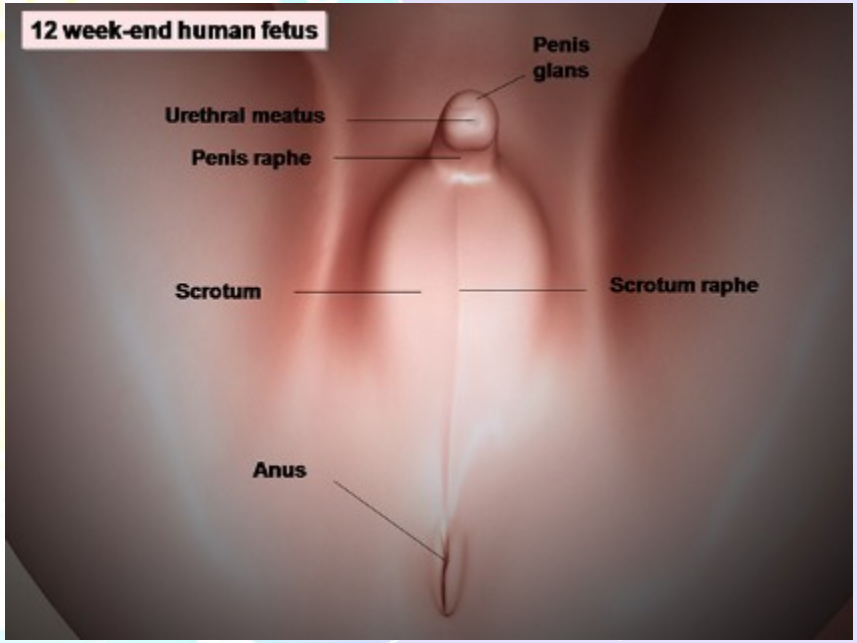
OSSIFICATION

- begins at the end of the embryonic period. Most primary ossification centers of skull, long bones and vertebrae appear by 12 weeks post fertilization.
- The epiphyses of long bones are still cartilaginous at birth.



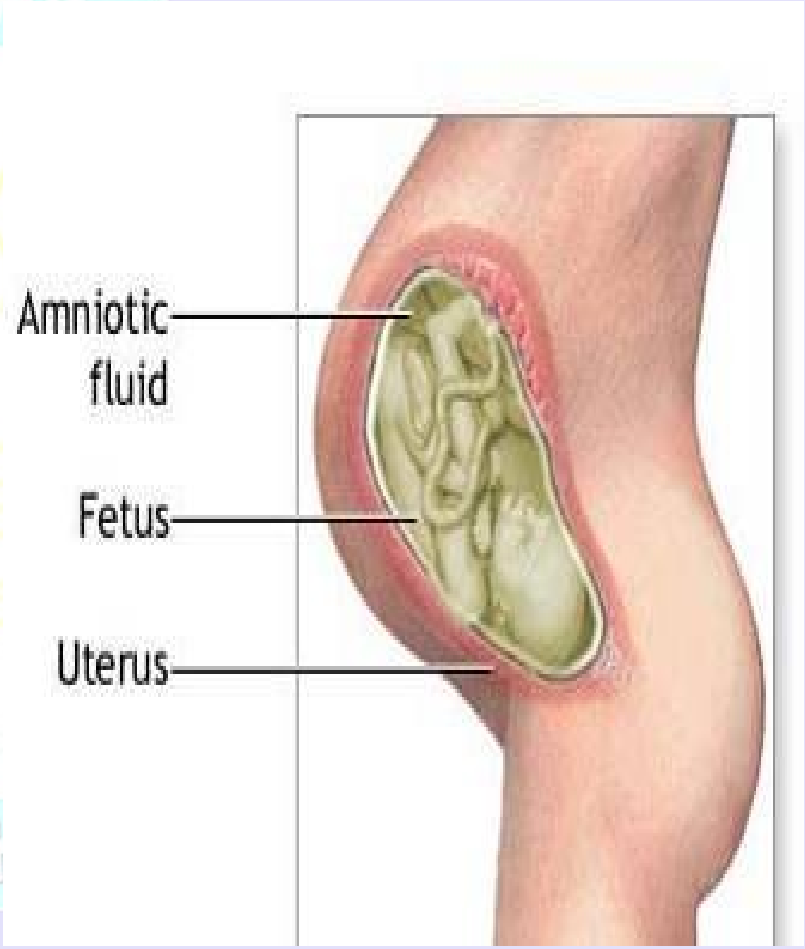
EXTERNAL GENITALIA

- The external genitalia sexually differentiate during weeks 9-12 post fertilization.



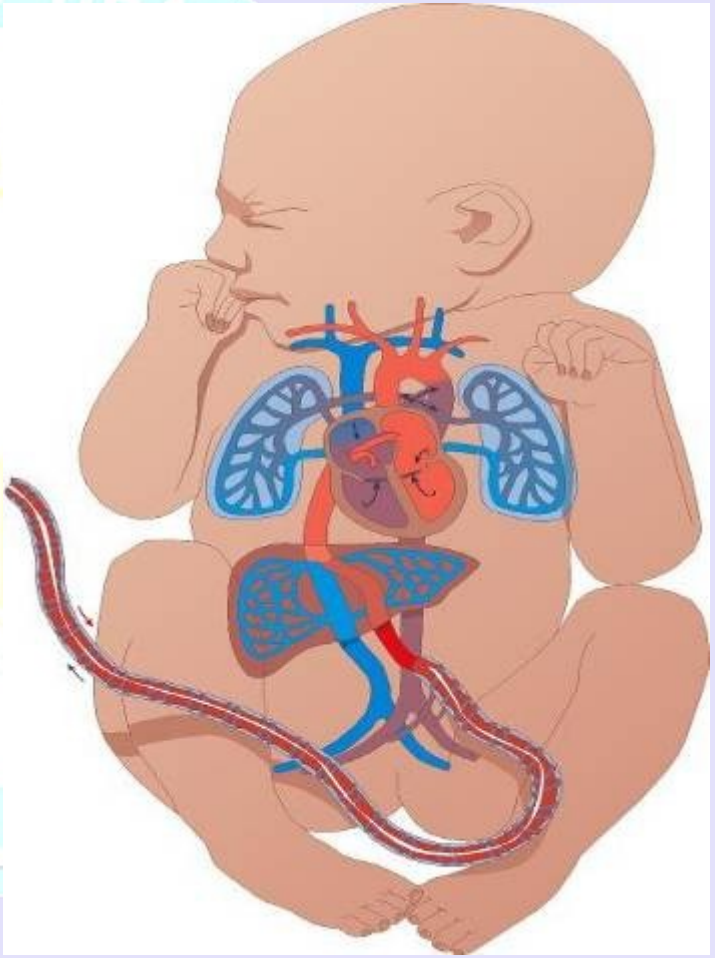
FETAL MOVEMENTS

- Mother first feels fetal movements at about 16-20 weeks post fertilization.
- This is the phenomenon known as "quickening".



LUNGS

- become capable of breathing air at the beginning of the 3rd trimester (about 25-26 weeks p.f.).
- when the lungs enter the saccular phase of development and begin to produce surfactant.
- Lung maturity is the primary determinant of viability.



HEMATOPOIESIS

- begins in the bone marrow in the second trimester.
- In the third trimester bone marrow replaces the liver and spleen as the primary site of hematopoiesis.

EMBRYONIC OR FETAL LOSS

- Embryonic or fetal loss before viability is termed a miscarriage or spontaneous abortion.
- Spontaneous abortion is very common (greater than 50% of pregnancies), especially in the first four weeks post-fertilization.
- Fetal loss after viability is termed stillbirth. Stillbirths account for 0.5-1% of births.

REFERENCES

Keith L. Moore Developing Human

8th Edition

THANK YOU